

encyclopedia became one of the most widely used treatises in European medical schools until the 16th century.

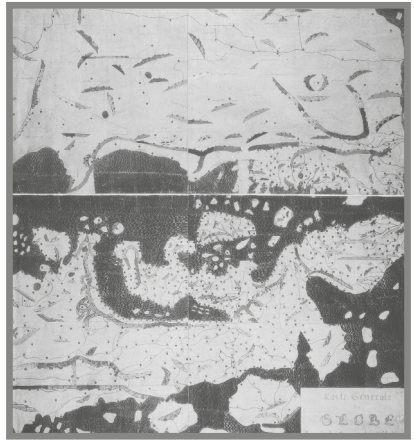
Royal patronage

This transmission of knowledge to the West mirrored the process by which the Islamic world had absorbed Greek learning during the great period of translation into Arabic in the 9th and 10th centuries. Noble and royal patrons played similar roles in both phases of the transmission. King Roger II of Sicily (which by 1091 had been reconquered from the Muslims) invited Arab scholar al-Idrisi to his court in 1138 with a commission to construct a map of the world based on Islamic geographical and cartographic works. The result, which took more than 15 years to complete, was by far the most accurate world map yet available to Europeans, and showed areas as far east as Korea. The map was accompanied by the *Book of Pleasant Journeys into Faraway Lands*, in which al-Idrisi's royal patron could have read of wondrous things such as cannibals in Borneo, and the gold trade in Ghana.

A tradition of learning

Roger's grandson Frederick II, Holy Roman Emperor from 1220 until 1250, continued his grandfather's tradition of sponsoring translations of Arabic texts. A remarkable polymath who knew at least four languages, Frederick so impressed his contemporaries with his learning that he became known as *Stupor Mundi* ("the Marvel of the World"). Among his protégés were the Scottish scholar Michael Scot, who translated key works of Aristotle on zoology, and the Pisan Leonardo Fibonacci, who had been sent by his merchant family to study mathematics at Bougie in Muslim North Africa. There Fibonacci learned of the decimal system, and in 1202 he published the *Book of Calculations*, the most detailed account yet seen in Europe of the Arabic system of numbering.

By the early 13th century, the Abbasid Empire had all but collapsed. The difficulties of ruling such a far-flung empire and the effects of a series of civil wars had led to key provinces such as Spain, Tunisia, and Egypt breaking away to make themselves independent. Even in Baghdad, where the



Roger II invited scholar al-Idrisi to create an accurate map of the known world in 1138. Al-Idrisi presented the planisphere, and an accompanying book, to his patron in 1154.

Abbasid caliphs clung on, they were only notionally sovereign. Real power was held by other dynasties such as the Shia Buyids, and, from 1055, the Seljuks, a Turkish group originating in central Asia. The final blow was dealt by the Mongols, who surged westward into the Islamic world in the early 13th century. In 1258, the Mongol Great Khan Möngke unleashed an army against Iraq, which laid siege to and then sacked Baghdad, inflicting an appalling massacre on its inhabitants. The last ruling Abbasid caliph al-Musta'sim was executed, and political and cultural leadership of the Islamic world passed first to the Mamluks in Cairo and then, after their conquest of Egypt in 1517, to the Ottoman Turks.

By this time Europeans had rediscovered Greek and Roman learning in almost every field of scholarship through the medium of Arabic texts. It had taken centuries for the new material to be absorbed, and a further wave of interest in classical manuscripts in the 15th century to spark the Renaissance in Europe. The House of Wisdom founded by the Abbasid caliphs had played a key role in ensuring the survival of Greek and Roman science in the Islamic world, allowing its transmission centuries later to Christian Europe. ■